TECHNICAL AND PRODUCT DOCUMENTATION · V1 · 24 OCT 2026

Property Token Platform

Fractional ownership of Pakistani real estate as regulated digital tokens. A private Ethereum-compatible chain holds the register, a public chain proves it was never rewritten, and every rupee moves only after two officers approve and three ledgers agree.

Demo live | Pre-pilot | ERC-3643 on Besu | CI green

4 QBFT validators, ~2 s finality	**ERC-3643** permissioned security tokens	**134** passing contract tests	**71 h** timelock on every change
30 append-only migrations	**0 gas** investors never hold crypto		

repo imgrasooldev/tokenize-property chain id 586586 anchor every 15 min

SECTION 1

Executive summary

What the platform is, who it is for, and where it stands today.

Each property is held by a trustee structure. Investors own units of that property as ERC-3643 security tokens on a permissioned blockchain operated by the platform. Rent is collected in rupees through a bank, split by holdings, taxed, and paid to investors' bank accounts. Every money or token action is proposed by one officer and approved by another, written to an append-only hash-chained ledger, and reconciled against the chain and the bank before anything is paid out.

Status, 24 October 2026. The core platform is built and running on a demo server with a live private chain. It is **not yet open to real investors**: KYC, bank, SMS and email providers are mock adapters, our own contracts have not had an external audit, and the legal wrapper (SECP sandbox, trustee agreement, SPV) is pending.

Which chain?

A private Hyperledger Besu network (Ethereum technology, QBFT consensus) with state hashes anchored to a public Ethereum Layer 2. Not Solana, not public Ethereum mainnet.

Who controls it?

Nobody alone. A 72-hour timelock owns every contract; multisig Safes for Admin, Trustee, Issuance and Operations sit behind it.

Where is the money?

In rupees, in trustee bank accounts. The chain records ownership and payout proofs; it never holds investor cash.

Biggest gaps?

External audit, real provider contracts, legal and SECP wrapper, and the final choice of public L2 for anchoring.

SECTION 2

The blockchain: what we use and why

A private chain as the system of record, a public chain as the public proof. Pick an option to see how it scores against what a regulated Pakistani issuer needs.

Public Ethereum mainnet

Investor cost	1 / 5
Privacy of holdings	1 / 5
Security-token standard	5 / 5
Regulatory control	3 / 5
Audit and tooling	5 / 5
Public proof	5 / 5

Every transfer costs gas in ETH, which a Pakistani investor cannot easily or legally hold. All holdings and wallets are public. Finality takes minutes.

Solana

Investor cost	4 / 5
Privacy of holdings	1 / 5
Security-token standard	1 / 5
Regulatory control	2 / 5
Audit and tooling	2 / 5
Public proof	5 / 5

Fast and cheap, but there is no mature permissioned security-token standard with built-in identity. Validators are not ours, so freezes and regulatory control depend on custom program design. Audit and formal-verification tooling is thinner.

Public Ethereum L2 only (Base, Arbitrum, Polygon)

Investor cost	3 / 5
Privacy of holdings	1 / 5
Security-token standard	5 / 5
Regulatory control	3 / 5
Audit and tooling	5 / 5
Public proof	5 / 5

Low cost and good tooling, but holdings and investor wallets are still public, and the platform still needs crypto on hand for every action. Kept as the anchoring layer instead.

Private Besu QBFT + public L2 anchor (chosen)

Investor cost	5 / 5
Privacy of holdings	5 / 5
Security-token standard	5 / 5
Regulatory control	5 / 5
Audit and tooling	5 / 5
Public proof	4 / 5

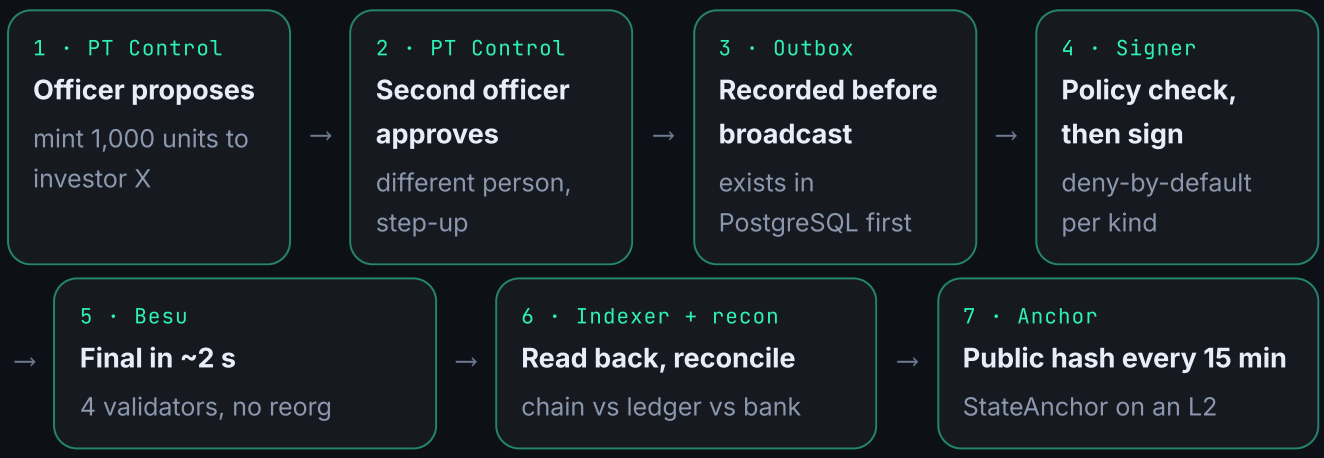
Zero gas for investors, private holdings, Ethereum tooling and audits, ERC-3643 unchanged, and a public hash trail that proves nobody rewrote history. Public proof is indirect (hashes, not transactions), which is the one trade-off.

The private chain

PROPERTY	VALUE
Software	Hyperledger Besu 26.9, the Linux Foundation enterprise Ethereum client
Consensus	QBFT, Byzantine fault tolerant. 4 validators today, tolerates 1 faulty node. Target 5 to 7 across platform, trustee and an independent party.

PROPERTY	VALUE
Chain ID	`586586`
Block time	About 2 seconds, immediate finality (no forks, no reorganisations)
Gas	Zero. The platform pays nothing per transaction and investors never need cryptocurrency.
Who writes	Only the signing service, through contracts owned by the timelock. Investors hold tokens in wallets bound to their verified identity.

How a transaction flows



1. **Officer proposes.** An operations maker fills the request in PT Control: for example mint 1,000 units to investor X against a completed subscription.
2. **Second officer approves.** A different officer with the checker duty reviews the exact calldata preview, passes a fresh hardware-key step-up, and approves. The approval digest is in the audit chain.
3. **Recorded before broadcast.** The command is written to the outbox table in PostgreSQL before anything is broadcast. If the process dies, the command still exists and is resumed exactly once.
4. **Policy check, then sign.** The signing service checks a deny-by-default policy for the action kind: allowed contract, allowed caller, daily limits, KYC status of the recipient, and that reconciliation currently allows payouts. Only then does it sign with a key from the encrypted keystore.
5. **Final in ~2 s.** Four Besu validators reach QBFT agreement in about two seconds. The block is final: there is no reorganisation and no fork.
6. **Read back, reconcile.** The indexer reads the Minted event back into the read model. Reconciliation compares chain supply and balances with the ledger and the bank; any mismatch freezes payouts until two operators clear it.
7. **Public hash every 15 min.** Within 15 minutes the anchor job writes the new private block hash and ledger head hash to the StateAnchor contract on the public chain. Anyone can later prove the private history matches.

Public anchoring

Every 15 minutes the platform writes a StateAnchor record to a public chain: the latest private block number, its hash, and the hashes of the ledger and distribution state. Anyone can later take a private block and prove it matches the public anchor, without any investor data leaving the private network.

Today: the anchor target is a local stand-in chain (Anvil, chain ID 586650) so the pipeline runs end to end in the demo; more than 180 anchors are recorded. **Production:** a public Ethereum L2 (Base, Arbitrum or Polygon PoS), an owner decision; a few US dollars a month at 15-minute cadence.

SECTION 3

Smart contracts

Audited standard code does the token logic. Our small, non-upgradeable guards are the only things allowed to touch it.

ERC-3643 CORE

ERC-3643 (T-REX 4.1) is the open standard for permissioned security tokens, used by regulated issuers worldwide. Unlike an ordinary ERC-20, every transfer is checked against an on-chain identity registry and compliance rules. We vendor the audited T-REX code unchanged.

Token, one per property

The units of ownership. Mint, burn, pause, freeze and forced transfer exist but can only be called by our guards.

ONCHAINID, one per investor

The investor's identity contract holding signed KYC and AML claims. Personal data stays off-chain; only claim hashes go on.

IdentityRegistry + Storage

Maps wallets to identities and countries. A wallet without a verified identity cannot receive tokens.

ModularCompliance

Runs rule modules on every transfer: CountryAllowModule (Pakistan, 586, only) and MaxBalanceModule (no identity above 10% of a property).

OUR GUARDS

GUARD	WHAT IT DOES	CALLER
IssuanceGuard	Only path to mint and burn. Supply can never exceed the trustee cap; one-time reference per subscription; no minting while paused.	Issuance Safe 2-of-3, at least one trustee signer
PauseGuard	Pause immediately; unpause only by the Admin Safe. A stolen on-call key can stop trading but never restart it.	Pauser key, Ops or Admin Safe to pause; Admin Safe to unpause
FreezeGuard	Freeze or unfreeze a wallet or part of a balance, tied to a compliance case ID.	Ops Safe
RecoveryGuard	Lost-wallet recovery and court-ordered forced transfers. Two steps, 48-hour delay, cancellable, works while paused.	Trustee Safe proposes and executes; Admin or Trustee cancels
RegistrarGuard	Registers KYC-approved investors, rate-limited on-chain to 50 per hour so a stolen automation key cannot flood the registry.	Registrar key in the signing service

Design rule. All guards are owned by the timelock, emit an event for every state change, have no payable functions, no delegatecall, no selfdestruct, and are not upgradeable.

REGISTRIES

DistributionRegistry

Each rent distribution is published as a Merkle root of (investor, amount). A 48-hour challenge window follows; the Trustee Safe can reject. After finalisation the SHA-256 of the bank payout file is recorded once.

DocumentRegistry

Append-only hashes of title deeds, trustee agreements, valuations and court orders. Records are never modified or deleted.

StateAnchor (public chain)

Receives the periodic hash of private chain state. Anchored block numbers must strictly increase.

KEYS AND CONTROL

TIMELOCK · 72 h

Owns every contract

Any change is visible for three days before it can execute. The Trustee Safe can cancel it. This includes the authorities that could upgrade tokens or identities.

SAFES · multisig

Admin 3-of-5 · Trustee 2-of-3 · Issuance 2-of-3 · Ops 2-of-3

Directors and CTO on hardware wallets in two cities; independent trustee officers; operations staff. Anything that creates or moves value needs one of these.

HOT KEYS

Pauser · Registrar · Claim issuer · Anchor

Can only do what is safe to do too often: pause, register an already KYC-approved investor, sign a claim, write a hash. A stolen hot key cannot move value.

HOW THEY ARE WRITTEN

- **1 · Specification first**

  Every contract starts from the written spec: a table of functions, callers and behaviours.

- **2 · Invariants I-1 to I-17**

  Each safety property is a numbered rule: supply never exceeds the trustee cap; a frozen address cannot send; the owner of every contract is the timelock.

- **3 · Foundry tests**

  Each invariant becomes a unit, fuzz or invariant test. 137 tests pass; a failing invariant blocks the merge.

- **4 · Static analysis and findings**

  Slither runs in CI with zero open findings. Every finding gets a regression test.

Audit status. The ERC-3643 base is audited by its maintainers. Our guards and registries have internal review and automated analysis only. An external audit is required before real investor funds.

SECTION 4

A property’s life on the platform

Eight stages. Click a stage, or use the arrow keys, to see who acts, what happens, and the control that makes it safe.

STAGE 1 · S-01 · SPONSOR · LEGAL · VALUATION · TECHNICAL · INVESTMENT COMMITTEE

Property onboarding

Title documents, valuation and technical report are reviewed by named duties; an investment committee with an independent member votes. Document hashes go to the DocumentRegistry.

CONTROL

A party to the file cannot approve its offering. Every reviewer duty is bound to an officer; maker-checker on each decision.

ON-CHAIN

DocumentRegistry: hashes of title deed, valuation, trustee agreement.

OFF-CHAIN

Property file, KYB of the sponsor, committee votes, audit chain.

STAGE 2 · Y-1 TO Y-6 · INVESTOR · KYC SERVICE · COMPLIANCE OFFICER

Investor KYC

CNIC check (NADRA Verisys in production), liveness and selfie, sanctions and PEP screening, wallet binding, ONCHAINID with KYC and AML claims, registration through RegistrarGuard.

CONTROL

Personal data sealed under keys outside the database; only peppered hashes and last four digits are readable. Sanctions hits need two officers.

ON-CHAIN

ONCHAINID claims, IdentityRegistry entry via RegistrarGuard (50/hour cap).

OFF-CHAIN

Sealed name and CNIC, screening results, status derived in the database.

STAGE 3 · S-02 · OPERATIONS MAKER AND CHECKER · INVESTORS · BANK

Offering

Terms approved; investors reserve first-come-first-served, get a unique payment code, pay in PKR by transfer or Raast. Exact credits confirm; wrong amounts go to suspense; a waitlist fills freed units; refunds and extensions are maker-checker.

CONTROL

Reserved plus paid never exceeds the hard cap, under a row lock. A payment code exists only for a reservation.

ON-CHAIN

Nothing yet: no token moves until completion.

OFF-CHAIN

Reservations, payment codes, bank credits, suspense, waitlist, refund requests.

STAGE 4 · I-2, I-11 · ISSUANCE SAFE · SIGNING SERVICE · INDEXER

Minting

Only when the offering completes. An issuance batch is approved, IssuanceGuard mints with a one-time reference per subscription, and the indexer confirms the balance before settlement is booked.

CONTROL

Supply can never exceed the trustee cap; each reference is used once; mint retries are approved separately.

ON-CHAIN

IssuanceGuard.mint per subscription; Minted events.

OFF-CHAIN

Issuance batch, settlement journal from the read model.

STAGE 5 · S-03 · OPERATIONS · EXTERNAL PROPERTY MANAGER · TAX OFFICER

Operations and rent

Leases, invoices, expenses and vendor payments are proposed and approved. Rent credits on the trustee escrow are matched to invoices. Withholding tax per investor from the Active Taxpayer List.

CONTROL

CNIC for tax sealed under a separate tax-only key; FBR export needs two tax officers and a one-time ticket. Seeded rates are placeholders.

ON-CHAIN

None until distribution.

OFF-CHAIN

Leases, invoices, expenses, escrow statements, ATL status, WHT schedule.

STAGE 6 · I-14 · OPERATIONS · TRUSTEE · BANK · RECONCILIATION

Distribution and payout

Net rent is split by holdings at a record block in integer paisa. The Merkle root is published on-chain; the trustee has 48 h to reject; then finalised. A bank payout file is approved by two operators and its SHA-256 recorded on-chain.

CONTROL

Payouts are blocked while reconciliation has a blocking finding; two operators lift the freeze after a passed run. Payee accounts have a 72 h cooling-off.

ON-CHAIN

DistributionRegistry: publish root, finalize, recordPayout(hash).

OFF-CHAIN

Statement per investor, bank file, held payouts, running balances.

STAGE 7 · S-04 · COMPLIANCE OFFICER · TRUSTEE OFFICER · OPERATIONS

Transfers (phase 0)

Case-by-case transfers between verified investors. Buyer pays into a per-case escrow; compliance and trustee approve; the trustee executes on-chain as a forced transfer while the token stays in transfer-closed mode; seller is paid after the chain confirms.

CONTROL

Delivery versus payment: money and tokens never move without each other. Approvers are different people, neither the opener nor the completer.

ON-CHAIN

RecoveryGuard ForcedTransfer propose, 48 h delay, execute.

OFF-CHAIN

Case, escrow, checks TR-10 to TR-13, refund path.

STAGE 8 · S-05 · OPERATIONS · LEGAL · TRUSTEE · INVESTORS

Exit

Property sale or wind-up. Proceeds land in escrow, a waterfall of payments is approved line by line, a final distribution is computed and published, redemption burns go through IssuanceGuard, and the case is closed.

Rent cycle in detail

CONTROL

Never pays out more than the pool holds; the trustee pre-checks the root; one live case per token.

ON-CHAIN

Final distribution root, IssuanceGuard.burn per redemption.

OFF-CHAIN

Exit case, waterfall lines, final statements, closure record.

Bank
Rent credit on escrow
matched to an invoice

→

Tax
Expenses and WHT
ATL status per investor

→

Ledger
Split at record block
largest remainder, integer paisa

→

Chain
Merkle root published
48 h trustee challenge

→

Recon gate
Chain = ledger = bank?
mismatch freezes payouts

→

Bank
Payout file, two operators
SHA-256 recorded on-chain

→

Investor
Paid to registered account
72 h cooling-off on changes

Lost wallets and succession

Through RecoveryGuard at any stage: the trustee proposes, a 48-hour delay runs, and the full balance moves to the re-verified wallet, even while the token is paused.

Notifications

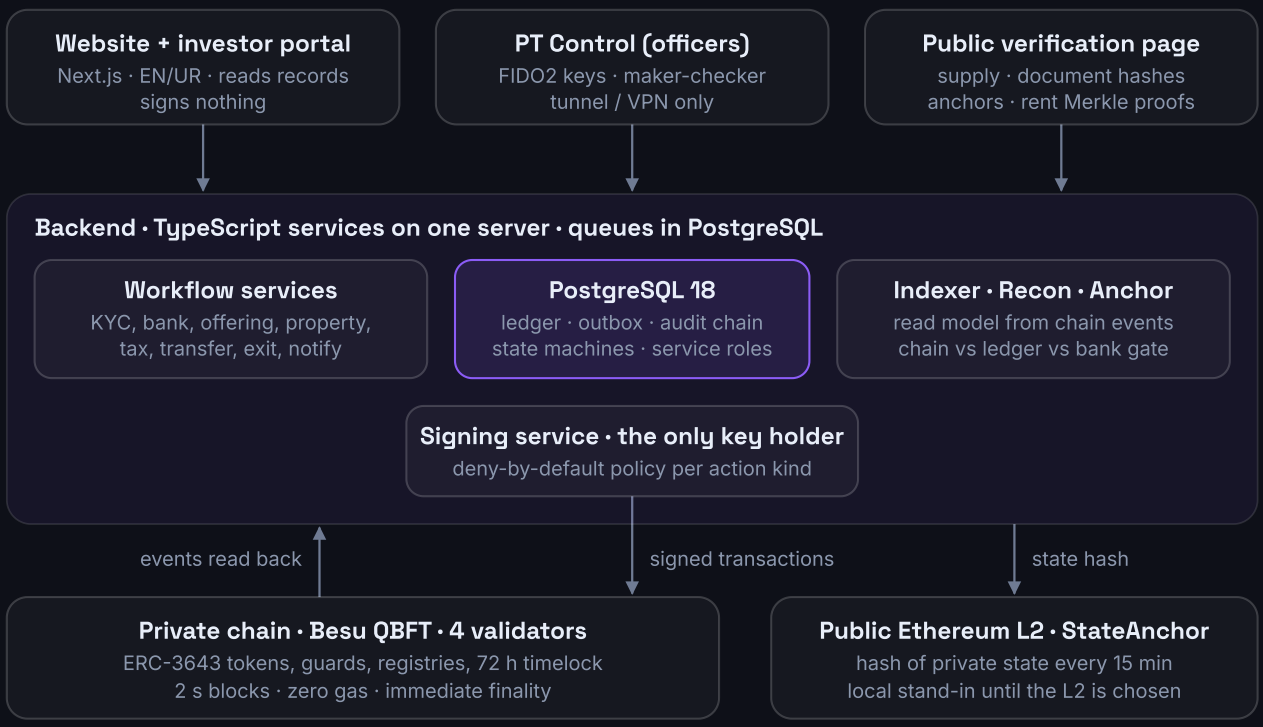
Raised inside the same database transaction as the event. Security notices cannot be switched off, carry no links, and go to the replaced contact as well as the new one.



SECTION 5

Platform architecture

A PostgreSQL-centred TypeScript backend, one component that holds keys, a private chain for tokens, and three Next.js front ends. Hover or tap a block to see its job and what it talks to.



> Portals write only to the database. The signing service is the single path from database to chain; the indexer reads the chain back so both are reconciled before any payout.

- **Website + investor portal.** Public website and investor portal. Server-rendered Next.js in English and Urdu. It reads the chain read model and database records and signs nothing: every investor action becomes a database command executed by the services.
- **PT Control (officers).** PT Control, the internal portal. FIDO2 hardware-key sign-in, database sessions with step-up, maker-checker approvals whose preview equals the signed calldata, and a hash-chained audit. Reachable only through a tunnel or VPN.
- **Public verification page.** Public verification page. Anyone can check token supply against the trustee register, verify a document hash, verify public anchors, and prove their own rent entitlement with a Merkle proof recomputed in the browser.
- **Workflow services.** Workflow services: KYC, bank, offering, property, tax, transfer, exit, recovery and notifications. Each has its own schema, its own database role, and a state machine enforced by triggers.
- **PostgreSQL 18.** PostgreSQL 18 is where every decision is first recorded: the double-entry hash-chained ledger, the outbox of chain commands, the audit chain, every workflow state machine, and 31 append-only checksum-protected migrations.
- **Indexer · Recon · Anchor.** Indexer, reconciliation and anchor. The indexer follows chain events into the read model; reconciliation compares chain, ledger and bank and freezes payouts on any mismatch; the anchor job publishes the state hash every 15 minutes.
- **Signing service · the only key holder.** The signing service is the only component with private keys. Every request is checked against a deny-by-default policy for its kind (mint, burn, freeze, record payout, transfer, identity, recovery), then signed from an encrypted keystore. HSM or MPC in production.
- **Private chain · Besu QBFT · 4 validators.** The private chain: Hyperledger Besu with QBFT consensus, four validators, two-second blocks, zero gas, immediate finality. It holds the ERC-3643 tokens, our guards and registries, all owned by a 72-hour timelock.
- **Public Ethereum L2 · StateAnchor.** The public anchor: a StateAnchor contract on an Ethereum Layer 2 receives the private block hash and ledger head hash every 15 minutes. Today a local stand-in chain;

the production L2 is an owner decision.

Database and security rules

Append-only migrations

31 checksum-protected migrations. A committed migration is never edited; fixes are new migrations.

State machines in triggers

Offering, exit, transfer and recovery transitions are enforced by the database, not only by application code.

Hash-chained audit

Every officer action is in a chained audit log with daily UTC digests; a verify function detects tampering.

Least-privilege roles

Each service has its own database role. The signer can read a bank file hash but cannot read bank tables.

Sealed personal data

Names, CNICs, phones and emails are AES-256-GCM sealed under keys outside the database, with separate keys for KYC, tax and notifications.

No single person, no single key

Hardware-key sign-in, step-up per action, a different approver, and a segregation-of-duties matrix with a verified digest.

SECTION 6

Portals, apps and roles

Four surfaces built today, three planned. Technically two Next.js apps plus the self-hosted Safe console.

SURFACE	USERS	WHAT IT DOES	STATUS
Public website	Everyone	How it works, live and upcoming properties with a rule-based score, fees, risk disclosure, FAQ, English and Urdu, grounded AI analyst that never gives advice, live chain pulse.	live on demo
Investor portal	Investors	Passkey sign-up and KYC, reserve and pay, portfolio, rent statements with Merkle proofs, payout account with cooling-off, transfer, exit and redemption requests, notifications.	built, gaps
Verification page	Anyone	Supply vs trustee register, document hashes, public anchors, verify your own rent entitlement.	live on demo
PT Control	Officers	Approvals, offerings, property files, operations and rent, tax, compliance cases and freezes, recovery, trustee register and distribution review, transfers, exits, officers and roles, audit viewer.	built, gaps
Safe console	Safe signers	Review and sign multisig transactions on hardware wallets.	production setup
Mobile app	Investors	Portal features plus biometrics, push, device binding (Flutter).	planned
Sponsor portal	Property owners	Submit a property, upload title documents, track due diligence.	later
Secondary market	Investors	Open buy and sell between verified investors.	after approval

Role explorer

Officers sign in with FIDO2 hardware keys only. Every role is read-only until a second officer

with the right duty approves. Pick a role to see what it can do, what it can never do, and which roles it can never be combined with.

ops_maker Live

CAN DO

- Prepare mint requests, distributions, payout batches
- Prepare leases, invoices, expenses, exits, transfers
- Open and complete cases

CAN NEVER DO

- Approve anything
- Approve own work

NEVER COMBINED WITH (8)

- admin
- auditor
- compliance_officer
- ops_checker
- platform_owner
- property_manager
- support_agent
- trustee_officer

Conflicts are refused by the database (admin.role_conflicts) whoever writes the grant, and the role matrix has a fixed digest that the audit verifies.

ops_checker Live

CAN DO

- Approve what makers prepared
- Approve settlements and refunds

CAN NEVER DO

- Prepare anything
- Approve own work

NEVER COMBINED WITH (8)

- admin
- auditor
- compliance_officer
- ops_maker
- platform_owner
- property_manager
- support_agent
- trustee_officer

Conflicts are refused by the database (admin.role_conflicts) whoever writes the grant, and the role matrix has a fixed digest that the audit verifies.

compliance_officer Live

CAN DO

- KYC decisions, sanctions review
- Freeze and unfreeze requests
- Case files, transfer compliance approval
- Masked support lookup

CAN NEVER DO

- Money or token actions
- Legal, tax or trustee decisions

NEVER COMBINED WITH (8)

- admin
- auditor
- ops_checker
- ops_maker
- platform_owner
- property_manager
- support_agent
- trustee_officer

Conflicts are refused by the database (admin.role_conflicts) whoever writes the grant, and the role matrix has a fixed digest that the audit verifies.

legal_officer Live

CAN DO

- Legal sign-offs on property files
- Exit decisions in legal capacity
- Recovery and succession review

CAN NEVER DO

- Money
- Compliance decisions

NEVER COMBINED WITH (5)

- auditor
- platform_owner
- property_manager
- support_agent
- trustee_officer

Conflicts are refused by the database (admin.role_conflicts) whoever writes the grant, and the role matrix has a fixed digest that the audit verifies.

tax_officer Live

CAN DO

- Tax rates
- ATL checks
- Two-officer FBR export

CAN NEVER DO

- Money
- Compliance

NEVER COMBINED WITH (5)

- auditor
- platform_owner
- property_manager
- support_agent
- trustee_officer

Conflicts are refused by the database (admin.role_conflicts) whoever writes the grant, and the role matrix has a fixed digest that the audit verifies.

trustee_officer Live

CAN DO

- Trustee register
- Co-sign issuance
- Reject distributions within 48 h
- Recovery requests, transfer execution

CAN NEVER DO

- Prepare anything
- Hold any platform role (independent trustee)

NEVER COMBINED WITH (9)

- auditor
- compliance_officer
- legal_officer
- ops_checker
- ops_maker
- platform_owner
- property_manager
- support_agent
- tax_officer

Conflicts are refused by the database (admin.role_conflicts) whoever writes the grant, and the role matrix has a fixed digest that the audit verifies.

auditor Live

CAN DO

- Read everything
- Export audit

CAN NEVER DO

- Any change

NEVER COMBINED WITH (10)

- admin
- compliance_officer
- legal_officer
- ops_checker
- ops_maker
- platform_owner
- property_manager
- support_agent
- tax_officer
- trustee_officer

Conflicts are refused by the database (admin.role_conflicts) whoever writes the grant, and the role matrix has a fixed digest that the audit verifies.

admin

Live

CAN DO

- Manage officers and roles, with a second admin

CAN NEVER DO

- Any business action

NEVER COMBINED WITH (6)

- auditor
- compliance_officer
- ops_checker
- ops_maker
- property_manager
- support_agent

Conflicts are refused by the database (admin.role_conflicts) whoever writes the grant, and the role matrix has a fixed digest that the audit verifies.

platform_owner

Drafted (0031)

CAN DO

- See everything: every read and audit permission
- Manage officers
- Place an emergency hold on payouts (recorded as a reconciliation freeze that two operators must lift)

CAN NEVER DO

- Prepare or approve money or token actions
- Compliance, legal, tax or trustee decisions

NEVER COMBINED WITH (9)

- auditor
- compliance_officer
- legal_officer
- ops_checker
- ops_maker
- property_manager
- support_agent
- tax_officer
- trustee_officer

Conflicts are refused by the database (admin.role_conflicts) whoever writes the grant, and the role matrix has a fixed digest that the audit verifies.

property_manager

Drafted (0031)

CAN DO

- Read its properties' operations
- Propose leases, invoices and expenses

CAN NEVER DO

- Approve anything
- See investor or officer data
- Export audit

NEVER COMBINED WITH (10)

- admin
- auditor
- compliance_officer
- legal_officer
- ops_checker
- ops_maker
- platform_owner
- support_agent
- tax_officer
- trustee_officer

Conflicts are refused by the database (admin.role_conflicts) whoever writes the grant, and the role matrix has a fixed digest that the audit verifies.

support_agent

Drafted (0031)

CAN DO

- Masked investor lookup by CNIC last four digits: status word, wallet, payout account last digits (audited)

CAN NEVER DO

- Decisions
- Exports
- Officer data

NEVER COMBINED WITH (10)

- admin
- auditor
- compliance_officer
- legal_officer
- ops_checker
- ops_maker
- platform_owner
- property_manager
- tax_officer
- trustee_officer

Conflicts are refused by the database (admin.role_conflicts) whoever writes the grant, and the role matrix has a fixed digest that the audit verifies.

SECTION 7

Feature inventory

Everything marked done is merged on main and running on the demo server with migrations 0001 to 0030 applied. Filter or search.

Blockchain done

- ERC-3643 suite with ONCHAINID
- Five guards, two registries, StateAnchor
- 72 h timelock ownership
- 4-validator Besu QBFT network
- Scripted deploy with invariant checks

Core backend done

- Hash-chained ledger with running balances
- Exactly-once outbox
- Signer policies for every action kind
- Indexer read model with verify
- Reconciliation with payout freeze
- Anchoring every 15 min

Workflows S-01 to S-07 done

- Property onboarding and committee votes
- Offering lifecycle with FCFS and refunds
- Operations, FBR withholding, distributions
- Transfers phase 0 with DvP escrow
- Exit, waterfall, redemption burns
- Recovery and succession; public verification

Front ends done

- Website EN/UR with real chain data, catalogue, search, analyst
- Passkeys, offerings, portfolio with rent Merkle proofs
- PT Control with 8 roles, approvals, compliance, trustee, tax, exit, transfers, audit
- End-to-end tests in CI, automated demo deploy

KYC, bank, notify done (mock providers)

- Sealed personal data, sanctions maker-checker, offboarding freeze
- Signed Raast webhooks, exact-match or suspense booking
- Payout accounts with 72 h cooling-off, held payouts
- Typed, link-free notifications, record-before-send

Platform roles partial

- platform_owner, property_manager, support_agent drafted in migration 0031
- Not applied, tested or deployed; support page not written

Rent on-chain publication partial

- Publish and finalize run through a script
- Signer kinds rent.publish and rent.finalize pending

Investor portal gaps partial

- Tax certificate download
- Demo personas need a KYC subject for notifications
- Offering withdrawal window, supplementary disclosure, document download

PT Control gaps partial

- CLI-only: bank statement import, suspense returns, follow-up batches, signer policy changes
- Session binding for S-01 maker actions
- Playwright for newest screens; enrol 14 demo officers

Providers and keys partial

- NADRA Verisys, liveness, sanctions, bank, Raast, SMS, email are mocks with real interfaces
- Anchor target is a local stand-in chain
- Encrypted file keystore; no HSM or MPC yet

Apps and portals not built

- Flutter mobile app and public API
- Sponsor portal
- Open secondary market (needs approval)

Investor features not built

- Tenant rent-collection link, auto-reinvest
- Complaints system, Roshan Digital Account onboarding
- Document vault, token-holder voting

Operations and assurance not built

- Monitoring and alerting stack
- Offsite encrypted backups (waiting on an offline GPG key)
- goAML regulatory reporting
- External audits, pen test, formal verification

SECTION 8

Testing, quality and the demo

Every push runs the full suite in GitHub Actions; a green run on main deploys the demo server automatically.

LAYER	WHAT IS TESTED	HOW
Smart contracts	137 Foundry unit, fuzz and invariant tests for I-1 to I-17; deployment checks; regression test per finding; Slither with zero open findings; 100% line coverage at the gate	forge test, Slither in CI
Backend	Per-package suites on scratch databases: ledger integrity, outbox exactly-once (property-tested), signer policies, indexer verify, reconciliation, KYC and bank rules, every state machine	Vitest
PT Control	207 unit and DB tests, 272-case HTTP permission matrix (every route vs every role), end-to-end sign-in and approvals with virtual FIDO2 keys	Vitest + Playwright
Website	67 tests and 11 end-to-end flows: sign-up, KYC, invest, portfolio, verify	Vitest + Playwright
Chain ops	50-of-50 flow checks on real blocks; validator outage and recovery drills	chain/flow.py

Security findings

Six recorded and fixed: three in vendored ERC-3643 code, three in ours (time-zone dependent audit digest, drifting running balances, a NULL comparison that blocked valid rows). Each has a regression test.

Demo server

One Ubuntu 24.04 VPS in Singapore: four Besu validators, PostgreSQL 18, all services, website on plain HTTP, PT Control behind an SSH tunnel. Verified 24 Oct 2026: CI and deploy green, chain past block 490,000, 180+ anchors.

Demo console

Scripted scenarios mint, run a rent cycle, and deliberately tamper with the ledger to show reconciliation catching it. Local test keys only; no real money or investors.

SECTION 9

Roadmap: before real investors

Five gates. Each is a hard stop: the next does not start until the previous passes and the owner signs it off.

Gate A

Finish the code

Platform roles 0031, rent publish/finalize signer kinds, remaining investor screens, CLI-only actions into PT Control, Playwright for new screens, demo officer enrolment.

Engineering

Gate B

Owner decisions

Public L2 for anchoring; offline GPG key for backups; retail investment limits; KYC retention; withholding rates confirmed by a tax adviser; branch protection and signed commits.

Owner

Gate C

Real integrations

NADRA Verisys, liveness vendor, sanctions and PEP screening, trustee bank Raast and payout API, SMS and email; HSM or MPC keys; monitoring; offsite encrypted backups.

Owner + engineering

Gate D

Assurance and legal

Two independent contract audits, penetration test, formal verification of the guards, GPL-3.0 opinion on ERC-3643, SECP sandbox application, trustee agreement and SPV, Shariah review if offered.

Auditors + counsel

Gate E

Pilot

Full rehearsal with the trustee on staging, then a limited first offering under SECP sandbox conditions.

Everyone

After the pilot: mobile app and public API, sponsor portal, open secondary market once approved, tenant rent-collection link, auto-reinvest, Roshan Digital Account onboarding, complaints system, document vault, token-holder voting, goAML reporting.

SECTION 10

Investor and regulator FAQ

The questions investors ask, with the answers the system actually supports.

Is this a cryptocurrency?

No. Tokens represent units of ownership in a specific property held by a trustee. They are priced and paid in rupees, cannot be sent to anonymous wallets, and have no value outside the platform's legal structure.

Which blockchain is it on? Ethereum? Solana?

A private, permissioned Ethereum-compatible network (Hyperledger Besu) run by the platform, with its state hashes published to a public Ethereum Layer 2. Ethereum technology, not public Ethereum mainnet, and not Solana.

Why not a public chain where I can see everything?

Public chains expose every holding and wallet and require everyone to hold crypto for fees. Our chain keeps holdings private while the public anchor proves the records were not altered. You can verify your own entitlement on the verification page.

Do I need a crypto wallet or to buy crypto?

No. Your wallet is created and bound to your verified identity during KYC. You never pay gas or hold cryptocurrency.

Who can hold the tokens?

Only investors who passed KYC (CNIC, liveness, sanctions screening) and are resident in Pakistan. A transfer to anyone else fails on-chain. No identity may hold more than 10% of a property.

What stops the platform from minting tokens for itself?

Minting goes only through IssuanceGuard, which refuses any supply above the units on the trustee's register, and only a multisig with a trustee signer can call it. Supply is publicly checkable against the register.

Can the platform freeze or take my tokens?

A freeze requires a compliance case and the Ops multisig, recorded on-chain with a case reference. A forced transfer requires the independent trustee, a 48-hour delay, and an evidence hash (for example a court order) in the DocumentRegistry.

What if I lose my wallet or pass away?

The trustee runs a recovery: identity is re-verified or succession documents are filed, a 48-hour delay runs, and the full balance moves to the new wallet.

How is rent paid?

Rent lands in the trustee's bank account; expenses and withholding tax are deducted; the split is published on-chain as a Merkle root; the trustee has 48 hours to object; then a bank file pays your registered account. The file's hash is recorded on-chain.

What if the bank, the ledger and the chain disagree?

Reconciliation runs the three-way check before every payout. Any mismatch freezes all payouts until two operators investigate and clear it.

Who holds the keys?

No individual. A 72-hour timelock owns every contract; multisig Safes of directors, trustee officers and operations staff sit behind it. Automated keys can only pause, register KYC-approved investors, or write anchors.

Has the code been audited?

The ERC-3643 base is audited upstream. Our own contracts have internal review, 137 automated tests and static analysis. Two external audits and a penetration test are scheduled before any real offering.

Where is my personal data?

Off-chain, sealed with encryption keys held outside the database. The chain holds only claim hashes. Even support staff see masked data only.

What is the regulatory status?

Pre-pilot. The platform is designed for the SECP regulatory sandbox; the application, trustee agreement and SPV structure are pending.

Can I sell my tokens?

Today, through case-by-case transfers to another verified investor, approved by compliance and executed by the trustee with delivery-versus-payment. An open secondary market comes only after regulatory approval.

What are the main risks?

Property and tenant risk, liquidity (no open market yet), regulatory outcome, and the platform's own operational risk. The technical controls reduce fraud and error risk; they do not remove investment risk.

SECTION 11

Glossary

Terms used across this document.

TERM	MEANING
ERC-3643 (T-REX)	Open standard for permissioned security tokens; every transfer is checked against identity and compliance rules

	checked against identity and compliance rules.
ONCHAINID	An investor's on-chain identity contract holding signed KYC and AML claims.
Hyperledger Besu	Enterprise Ethereum client used to run the private chain.
QBFT	Byzantine fault tolerant consensus for private chains; immediate finality, no forks.
Layer 2 (L2)	A public network settled on Ethereum (Base, Arbitrum, Polygon). Used only for anchoring.
StateAnchor	Our public contract storing periodic hashes of the private chain.
Guard	A small contract of ours that is the only allowed caller of a sensitive token function.
Timelock	A contract that delays every ownership action by at least 72 hours so it can be reviewed or cancelled.
Safe (multisig)	A wallet that needs several signers (for example 3 of 5) to act.
Trustee	The independent party that holds the property and co-signs issuance, distributions and recoveries.
Maker-checker	One officer prepares, a different officer approves.
Segregation of duties	Roles that cannot be held by the same person, enforced in the database.
Outbox	A command is recorded in the database before it is sent to the chain, so nothing is lost or sent twice.
Reconciliation	The three-way comparison of chain, ledger and bank that gates payouts.
Merkle root	One hash committing to every (investor, amount) line of a distribution; each investor can prove their own line.
DvP	Delivery versus payment: tokens and money move together or not at all.
KYC / AML / PEP	Know your customer, anti money laundering, politically exposed person checks.
ATL / WHT	FBR Active Taxpayer List; withholding tax deducted from rent.
Raast	Pakistan's instant payment system, used for subscriptions and payouts.
Passkey / FIDO2	Phishing-resistant sign-in with a hardware key or device biometrics.

TERM	MEANING
Paisa	One hundredth of a rupee; all money is stored as integer paisa.
SECP	Securities and Exchange Commission of Pakistan.

Property Token Platform documentation · prepared 24 October 2026 from the repository's specs, ADRs and status log. Test counts, block height and anchor counts were read from CI and the demo server on that date.